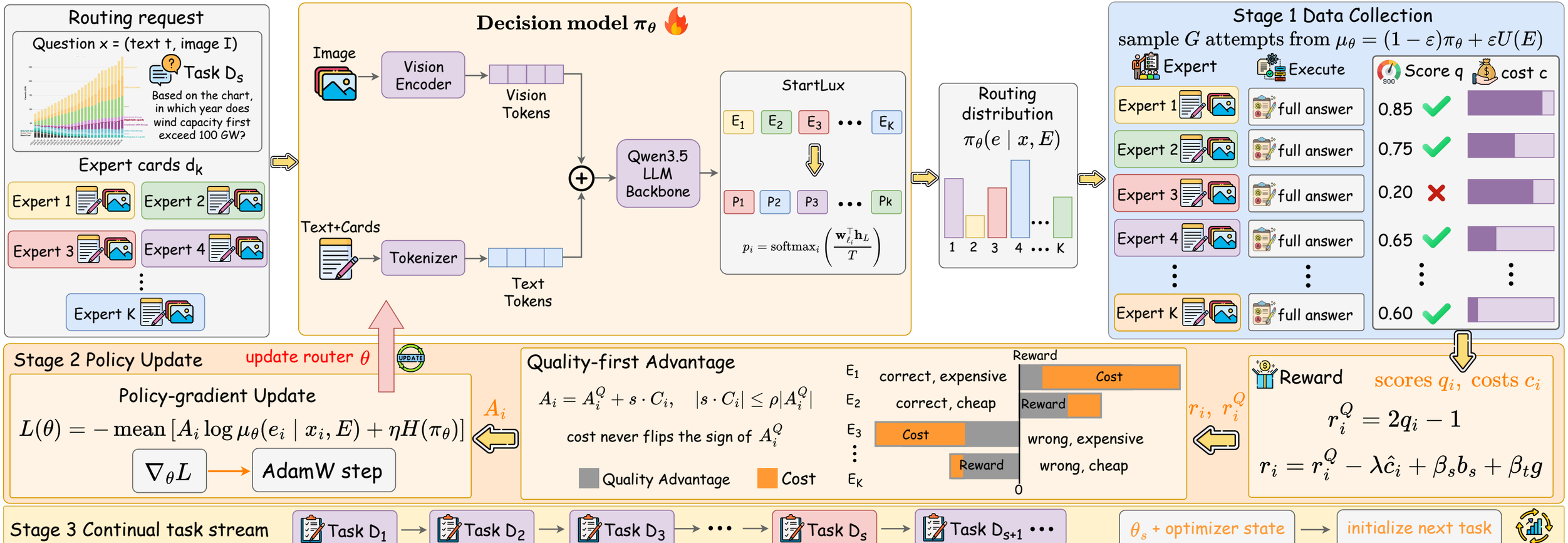

Routing request
Question x = (text t, image I)
Task D_s
Based on the chart, in which year does wind capacity first exceed 100 GW?
Expert cards d_k
Expert 1
Expert 2
Expert 3
Expert 4
Expert K
Decision model π_θ
Image
Vision Encoder
Vision Tokens
Text+Cards
Tokenizer
Text Tokens
Qwen3.5 LLM Backbone
StartLux
E_1 E_2 E_3 ••• E_K
P1 P2 P3 ••• Pk
$p_i = \mathrm{softmax}_i\left(\frac{\mathbf{w}_{\ell_i}^\top \mathbf{h}_L}{T}\right)$
Routing distribution
$\pi_\theta(e \mid x, E)$
1 2 3 4 ••• K
Stage 1 Data Collection
sample G attempts from $\mu_\theta = (1-\varepsilon)\pi_\theta + \varepsilon U(E)$
Expert
Execute
Score q
cost c
full answer
0.85
0.75
0.20
0.65
0.60
Stage 2 Policy Update
update router θ
UPDATE
Policy-gradient Update
$L(\theta) = -\,\mathrm{mean}\,[A_i \log \mu_\theta(e_i \mid x_i, E) + \eta H(\pi_\theta)]$
$\nabla_\theta L$
AdamW step
A_i
Quality-first Advantage
$A_i = A_i^Q + s \cdot C_i, \quad |s \cdot C_i| \le \rho |A_i^Q|$
cost never flips the sign of A_i^Q
Quality Advantage
Cost
Reward
correct, expensive
correct, cheap
wrong, expensive
wrong, cheap
0
$r_i,\ r_i^Q$
Reward
scores q_i, costs c_i
$r_i^Q = 2q_i - 1$
$r_i = r_i^Q - \lambda \hat{c}_i + \beta_s b_s + \beta_t g$
Stage 3 Continual task stream
Task D_1
Task D_2
Task D_3
Task D_s
Task D_{s+1}
θ_s + optimizer state
initialize next task